

# LENA LIEBICH

HARVARD BUSINESS SCHOOL AI INSTITUTE, DIGITAL RESKILLING LAB

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## Academic Appointments

Harvard Business School AI Institute

Postdoctoral Fellow, Digital Reskilling Lab

Boston, MA

2026 – present

Leibniz Institute for Financial Research SAFE

Research Affiliate

Frankfurt am Main, Germany

2026 – present

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## Research Interests

Human-AI interaction, experimental and behavioral economics, organizational economics

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## Publications

### [1] Knowing (Not) to Know: Explainable Artificial Intelligence and Human Metacognition

von Zahn, M., Liebich, L., Jussupow, E., Hinz, O., & Bauer, K. (2025)

*Forthcoming in Information Systems Research, published online in Articles in Advance.*

*Abstract:* The use of explainable artificial intelligence (XAI) to render black-box artificial intelligence (AI) more interpretable to humans is gaining practical relevance. Prior research has shown that XAI can influence how humans “think”. Yet, little is known about whether XAI also affects how people “think about their thinking” (i.e., their metacognitive processes). We address this gap by investigating whether XAI affects metacognitive processes that guide human confidence judgments about their ability to perform a task and thereby, their decision whether to delegate the task to AI. We conducted two incentivized experiments in which domain experts repeatedly performed prediction tasks, with the option to delegate each task to an AI. We exogenously varied whether participants initially received explanations about AI’s overall prediction logic. We find that AI explanations improve how well humans understand their own ability to perform the task (metacognitive accuracy). This improvement causally increases both the frequency and effectiveness of human-to-AI delegation. Additional analyses show that these effects primarily occur when explanations reveal to humans that AI’s prediction logic diverges from their own, leading to a systematic reduction of overconfidence. Our findings highlight metacognitive processes as a central, previously overlooked channel through which AI explanations can influence human-AI collaboration. We discuss practical implications of our results for organizations implementing XAI to comply with regulatory transparency requirements as, for example, outlined in the European Union Artificial Intelligence Act.

PAPER

DOI

### [2] Can GPT Mimic Human Preferences? An Empirical and Structural Investigation

Bauer, K., Liebich, L., & Kosfeld, M. (2025)

*Proceedings of the European Conference on Information Systems (ECIS 2025)*

*Abstract:* We examine the depth of large language models’ (LLMs) potential to serve as tools for simulating human decision-making processes. Using GPT-4 Turbo in a sequential prisoner’s dilemma, we find significant discrepancies between LLM-generated and human behaviors: GPT consistently exhibits a higher propensity to cooperate and forms overly optimistic expectations of human cooperation, leading to outcomes that diverge from observed human decision-making patterns. Yet, GPT’s decisions do not appear random; they align with a formal model of human preferences for fairness and efficiency that explains the behavior of most human participants in similar settings. We find a striking

contrast: while GPT may fall short as accurate surrogate for human decisions at the surface level, the model nonetheless exhibits a capacity to reflect the underlying preferences that drive human behavior in strategic contexts. Our findings have implications for LLMs' potential to serve as a managerial or research tool in simulating human behaviors.

PAPER

## Work in Progress

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### [3] Responsive Advisors, Indifferent Clients: Evidence from a Field Experiment on the Role of Explainable AI in Financial Advice (Working Title)

Liebich, L., Bauer, K., Yang, C., Li, X. & Hinz, O. (2026)

### [4] Personalized Communication and Repayment (Working Title)

Hett, F., Gill, A., Laudenbach, C., Liebich, L. & Siegel, S. (2026)

## Policy Work (Pre-PhD)

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### [5] Unconventionally Green: Monetary Policy between Engagement and Conflicting Goals

Liebich, L., Nöh, L., Rutkowski, F., & Schwarz, M. (2023)

*Review of Economics*, 74(1), 53–77.

*Abstract:* We study the potential role of the ECB in supporting the transition to a low-carbon economy by decarbonizing the corporate sector purchase programme (CSPP). We demonstrate that the carbon intensity of CSPP purchases is determined by three main factors: First, by the CSPP-eligibility criteria, as these tend to exclude bonds from low-emission sectors. Second, by the underlying structure of the bond market, as this tends to be skewed towards carbon-intensive sectors. Third, among the eligible bonds, the ECB tends to select those from relatively emission-intensive sectors. To decarbonize the CSPP, the ECB can theoretically act along these three lines: (i) adjust the CSPP-eligibility criteria to expand the range of eligible low-carbon assets, (ii), revise the principle of market neutrality to tilt the CSPP portfolio towards low-carbon companies, or, (iii), purchase so far neglected low-carbon bonds within the current eligibility and market neutrality framework. As all approaches have either very limited effects or are associated with significant theoretical and practical concerns, we conclude that their contribution to the success of active green monetary policy is not guaranteed, while at the same time risks arise for a monetary policy that targets price stability.

PAPER

DOI

### [6] Current Developments in Green Finance

Liebich, L., Nöh, L., Rutkowski, F., & Schwarz, M. (2020)

*German Council of Economic Experts, Working Paper 05/2020.*

*Abstract:* The transformation of economies towards significantly reduced CO<sub>2</sub> consumption raises high investment and capital requirements. Financial and capital markets can help to mobilize the necessary funds for global investment needs and to steer capital towards sustainable investments. Moreover, potential disruptive impacts of climate change on the financial system have started to become more apparent recently and require central banks, regulators and supervisors to take a conscious look at the risks and opportunities of climate change for financial intermediaries and markets. This article offers a comprehensive discussion on how green finance has been evolving thus far and explores the opportunities and key developments ahead with particular emphasis on four selected highly topical issues: 1) the introduction of German green government bonds, 2) obstacles to the correct pricing of climate-related risks, 3) the EU taxonomy that has recently been put forward to develop a uniform classification of sustainable economic activities as well as 4) the role of central banks in fostering the transition to a low-carbon economy.

PAPER

## Education

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### Goethe University Frankfurt (GSEFM)

PhD in Economics, *summa cum laude*

Dissertation: *Essays on Human-AI Interaction and Household Finance*

Committee: Christine Laudenbach, Michael Kosfeld, Guido Friebe, Florian Hett

Frankfurt am Main, Germany

2020 – 2026

### University of Melbourne

Visiting Scholar, Centre for Brain, Mind and Markets

Two-month research stay at the invitation of Carsten Murawski.

Melbourne, Australia

2025 – 2026

### CUHK Business School

Visiting Scholar, Department of Decisions, Operations and Technology

Two-month research stay at the invitation of Jenny Jin.

Hong Kong SAR

2025

### Goethe University Frankfurt (GSEFM)

MSc in Quantitative Economics

Focus: applied econometrics, household finance, experimental finance.

Frankfurt am Main, Germany

2020 – 2025

### University of Bonn

BSc in Economics

Focus: experimental economics, game theory, contract theory, applied microeconomics.

Bonn, Germany

2016 – 2020

## Teaching

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### Gutenberg School of Business, JGU Mainz

Teaching Assistant, *Machine Learning for Management Decisions*

MBA course with Kevin Bauer.

Mainz, Germany

2025

### University of Bonn

Teaching Assistant, *Financial Markets and Institutions*

BSc Economics course with Isabel Schnabel.

Bonn, Germany

2019

## Awards & Scholarships

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### Award for Excellence in Teaching

Joint with Kevin Bauer, for the MBA course *Machine Learning for Management Decisions*, Gutenberg School of Business, JGU Mainz

2026

### Best Discussion Award

9th SAFE Household Finance Workshop, sponsored by Behavioural Finance e.V. (€1,000)

2025

### First-Year PhD Scholarship

Leibniz Institute for Financial Research SAFE

2020 – 2021

## Conferences, Seminars and Invited Talks

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- 4th Workshop on Behavioral, Digital and Financial Economics, JGU Mainz (presentation)
- Digital Economy Workshop (presentation)
- SAFE Brown Bag Seminar (presentation)

2026

- SAFE Junior Scholars Workshop: The New World of Corporate Boards (presentation) 2025
- 3rd Workshop on Behavioral, Digital and Financial Economics, JGU Mainz (presentation)
- University of Oxford, Faculty of Law (presentation)
- 9th SAFE Household Finance Workshop (discussion)
- CUHK Business School (presentation)
- 38th Australasian Finance and Banking Conference, UNSW Business School (presentation)
- 2nd Workshop on Behavioral, Digital and Financial Economics, JGU Mainz (presentation) 2024
- SAFE Brown Bag Seminar (presentation)
- 9th Spring School in Behavioral Economics, UCSD Rady School of Management (poster)
- Machine+Behavior Conference, MPI / Toulouse School of Economics / University of British Columbia (poster)
- 8th SAFE Household Finance Workshop (discussion)
- EconTribute Workshop on Human-AI Collaboration, University of Cologne (lightning talk)
- 1st Workshop on Behavioral, Digital and Financial Economics, JGU Mainz (presentation) 2023
- PhD course *Subjective Beliefs, Attention and Economic Behavior*, University of Copenhagen (poster)
- 1st Bonn-Frankfurt-Mannheim PhD Conference (presentation)
- Workshop on Interactions of Humans and Algorithms, TU Berlin (poster)
- SAFE Brown Bag Seminar (presentation)
- Society for Experimental Finance Summer School, University of Sofia
- 7th SAFE Household Finance Workshop (presentation)
- CORA Conference on Robots and Automation, Goethe University (presentation)
- ZEW / University of Mannheim Experimental Seminar (presentation)
- 6th SAFE Household Finance Workshop (discussion) 2022

## Academic Service

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| <b>Leibniz Institute for Financial Research SAFE</b><br>Supervision of Master's and Bachelor's theses, Department of Household Finance  | 2022 – 2026 |
| <b>Leibniz Institute for Financial Research SAFE</b><br>Representative of PhD Researchers                                               | 2022 – 2024 |
| <b>Graduate School of Economics, Finance and Management (GSEFM)</b><br>Member of Student Representatives                                | 2020 – 2024 |
| <b>Goethe University Frankfurt</b><br>Organizing Committee, 1st Bonn-Frankfurt-Mannheim PhD Conference<br>Keynote by Ulrike Malmendier. | 2023        |
| <b>European Conference on Information Systems (ECIS)</b><br>Referee                                                                     | 2025        |

## Professional Experience

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### Academic

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| <b>Leibniz Institute for Financial Research SAFE</b><br>Junior Researcher / Doctoral Student, Department of Household Finance                                            | Frankfurt am Main, Germany<br>2022 – 2025 |
| <b>European Central Bank</b><br>Academic Researcher, DG Macropprudential Policy and Financial Stability<br>Research on market-based indicators of climate systemic risk. | Frankfurt am Main, Germany<br>2022 – 2023 |

## European Central Bank

Trainee, DG Macprudential Policy and Financial Stability

Empirical analysis of euro-area banks' exposure to climate-related systemic risk following the CRISK methodology of Jung, Berner and Engle.

Frankfurt am Main, Germany

2022

## Leibniz Institute for Financial Research SAFE

Student Research Assistant

Data analysis and literature search supporting the faculty's research projects.

Frankfurt am Main, Germany

2021 – 2022

## German Council of Economic Experts (GCEE)

Research Intern, Scientific Staff

Empirical analysis of the carbon intensity of the ECB's bond purchases under the Corporate Sector Purchase Programme, contributing to GCEE working papers 05/2020 and 05/2021.

Wiesbaden, Germany

2020

## Non-Academic

### Consulting, media and language tutoring

2015 – 2020

- Consulting Assistant and Consulting Intern, CPC Unternehmensmanagement AG, Bonn & Frankfurt am Main
- Student Assistant, Editorial Office *Documentaries*, phoenix / ZDF, Bonn
- German Language Tutor, German School Abroad Bilbao, Spain

## Affiliations

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- Harvard Business School AI Institute
- Leibniz Institute for Financial Research SAFE (Research Affiliate)
- Behavioural Finance e.V. (Member)
- Mentoring Hessen, ProCareer.Doc Program (2024 cohort)

## Personal

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|-------------|-----------------------------------------------------|
| Software    | Stata, Python, LaTeX, Git, oTree, MS Office         |
| Languages   | German (native), English (fluent), Spanish (fluent) |
| Citizenship | German                                              |

## References

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Available on request.